

CLIENT DELIVERABLE

Auto-Scheduling Optimizer — Specification Document

Systack Documentation

ROLE	COLOR	HEX
Headers, CTAs	Navy	 #001a2d
Navy Light	Navy Light	 #002845
Secondary accents	Teal	 #007da9
Primary buttons, links	Cyan	 #00a1db
Gradients	Cyan Bright	 #00c5e0
Backgrounds	Gray 50	 #f8fafc
Cards	Gray 100	 #f1f5f9
Borders	Gray 200	 #e2e8f0
Muted text	Gray 400	 #94a3b8
Body text	Gray 600	 #475569
Headings	Gray 800	 #1e293b
Success	Green	 #22c55e
Error	Red	 #ef4444
Accent highlights	Purple	 #8b5cf6

Auto-Scheduling Optimizer — Specification Document

Automation ID: scheduling-optimizer

Version: 1.0

Status: Draft — Specification

Date: 2026-06-16

Builder: SOL / Assembly

Brand Reference

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1. Executive Summary

What It Does

Detects gaps in the schedule (30-min holes, dead zones) and automatically creates discount slots. Pushes availability to past customers via SMS/email to fill empty time.

Business Value

METRIC	BEFORE	AFTER (TARGET)
Schedule utilization	~70%	> 90%
Revenue from discount slots	\$0	\$200–500/week
Dead time	2–3 hrs/day	< 30 min/day

2. System Architecture

Gap Detection

```
Calendar Analysis (daily at 6 PM for tomorrow)
  → Scan all slots for next 48h
    → Identify gaps:
      - 30+ min hole between bookings
      - Full hour+ with no bookings
      - Day-before empty slots
    → Create discount slot:
      - 20% off for 30-min gaps
      - 30% off for 60-min gaps
      - 40% off for empty day-before slots
    → Push to past customers:
      - Email: "Last-minute availability – 20% off!"
      - SMS: "Flash slot open today at 2 PM – 30% off"
```

Components

COMPONENT	TECHNOLOGY	PURPOSE
Calendar Scanner	n8n Cron + Code	Detects gaps
Discount Engine	n8n Code Node	Creates discount slots
Notification	n8n Email/SMS	Pushes to customers

COMPONENT	TECHNOLOGY	PURPOSE
Booking Handler	n8n Webhook	Fills discount slots

3. Implementation Plan

PHASE	TASK	EST. EFFORT
1	Calendar integration	3 hours
2	Gap detection algorithm	2 hours
3	Discount slot creation	2 hours
4	Customer notification	2 hours
5	Fill-rate tracking	1 hour
Total		~10 hours

4. Revenue Impact Model

ASSUMPTION	VALUE
Daily dead hours	2
Slots created/week	10
Fill rate	40% (4 filled)
Avg discounted booking	\$40
Weekly additional revenue	\$160
Monthly	\$640
Annual	\$7,680

Specification document — not yet built.
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